An Introduction to Sentiment Analysis Using NLP and ML

Would you like to understand how Google uses NLP and ML for creating brilliant apps such as Google Translate? Would you like to build the 'next big thing' in the natural language understanding space? Then this article is for you. It introduces you to sentiment analysis of text based data with a case study, which will help you get started with building your own language understanding models.

We live in the age of Big Data. There is no shortage of text based data. If anything, one can argue that we have too much of it. Everything, from tweets on Twitter to comments on YouTube to customer reviews on Amazon, can be considered as text based data. As human beings, it's easy for us to read a piece of text in a language that we are familiar with, and understand what emotions are being expressed and what specifications are being shared. However, it's not as simple for a machine such as a computer to do the same.

Natural language understanding

While computers have been able to deconstruct and analyse syntactic rules of a language for decades, with software such as MS Word being able to recognise spelling and punctuation errors, semantic understanding of a language is a whole different ball game. This is where natural language processing (NLP) and machine learning come into the picture. For decades, researchers have been working hard to make machines that are able to understand what is being expressed and the underlying emotions that are being exhibited in a human language. Although the techniques for creating such a technology have been known for quite a while, i.e., smart algorithms that can learn from data, what was lacking was the real-time 'data' required to train the algorithms. Additionally, there was an element of computational complexity that required smarter devices with faster processing speed to be able to analyse a piece of text in real-time and share the results instantly.

In 2022, we have the data, the speed, and the algorithms to finally make this happen. The past decade (2010 onwards) has been monumental for natural language processing. With the advent of cloud technology, coupled with Big Data frameworks, scientists have made immense strides in achieving 'natural language understanding'. This has been achieved by using natural language processing in conjunction with smart machine learning algorithms that can work with both structured and unstructured data.

Sentiment analysis of text

In the field of natural language processing of textual data, sentiment analysis is the process of understanding the sentiments being expressed in a piece of text. As humans, we communicate both the facts as well as our emotions relating to it by the way we structure a sentence and the words that we use. This is a complex process that, albeit seems simple to us, is not as easy for a computer to deconstruct and analyse. Sentiment analysis of text requires using sophisticated natural language processing techniques coupled with advanced machine learning algorithms that have the ability to learn from structured as well as unstructured data.